

PCB Design and Transformer-less Power Supply Notes

Documentation

December 16, 2025

1 Transformer-less Power Supply Project

Overview

Objective: DC power supply – 220V AC to 5V DC.

Components Used

- LM7805 Voltage Regulator (Linear)
- Diode 1N4007 $\times$ 4
- Caps (Capacitors) Polarised & Unpolarised $\times$ 1
- 2 $\times$ Zener diode

Circuit Analysis & Component Functions

- **Input Stage (Capacitor & Resistor):**
 - Specification: $225K/2.2\mu F$ (Note: This is the code by which this capacitor is known in the market).
 - Diagram Note: The resistor is placed in parallel with the capacitor at the I/P.
 - Function: Drops down the incoming voltage. The resistor discharges the capacitor.
- **Zener Diode:** Approx 6V.
- **LM7805 Regulator:**
 - Handles 50 to 60V DC.
 - Whatever is incoming from the behind circuit will be a pulsating D.C.
 - **Function:** The LM7805 converts the I/P (Input) to stable DC O/P (Output).
- **Filtering Capacitors:**
 - The capacitors filter out the pulsating voltage & passes the O/P to LM7805.
 - The polarised capacitor after 7805 is used to stabilize the voltage.
- **Net Labelling:** Used to know where is what voltage.

ERC (Electrical Rules Check)

- **Error Found:** "Input Power not driven by any O/P power".
- **Solution:** Usage of a simple power symbol from Power Source tab. A **power-flag** symbol is used to solve this.
- **Result:** ERC done woohoo!! no error.

3D Modeling in PCB Software

- **How to add any 3D models:**
 1. Click on the item that has not the 3D model.
 2. Download 3D model of that.
 3. Click E and then select that file.
- **Problem Faced:** I couldn't get the correct terminal 3D model. I got the 45° degree one and it was not in proper orientation for which I had to rewire the terminals.
- **Result:** I got the entire 3D model of all the components.

2 PCB Design Workflow

